

HIGH PERFORMANCE COMPUTING

Tugas 1



Disusun oleh:

Wilson Angelie Tan - 140810230024

UNIVERSITAS PADJADJARAN

FAKULTAS MATEMATIKA DAN ILMU PENGETAHUAN ALAM

Program Studi S-1 Teknik Informatika

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Source Code: <https://github.com/WilsonnnTan/hpc-assignment/tree/master/assignment-1>

Config bisa diubah pada bagian script:

```
#!/bin/bash
#
# Implementation of basic full and differential backup script
# This script doesnt support resuming an existing backup
# Everytime this script run, it will generate a new backup folder
# This script must be executed on dir that had $WATCHED_FOLDER

# Global and Config Vars
WATCHED_FOLDER="$(pwd)/hpc-dir"
BACKUP_NAME="your-project-name"
BACKUP_FOLDER="$(pwd)/backup/$BACKUP_NAME"
BACKUP_INTERVAL=1 # minute
FULL_BACKUP_FOLDER=""

if [[ ! -d "$WATCHED_FOLDER" ]]; then
    echo "Missing folder: WATCHED FOLDER must exist"
    exit 1
fi

if [[ -d "$BACKUP_FOLDER" ]] then
    echo "Removing existing BACKUP_FOLDER to avoid dup/corrupted backup"
    rm -r "$BACKUP_FOLDER"
fi

# helper
get_timestamp() {
    timestamp=$(date -u +%Y-%m-%d_%H-%M-%S_UTC)
    echo "$timestamp"
}
```

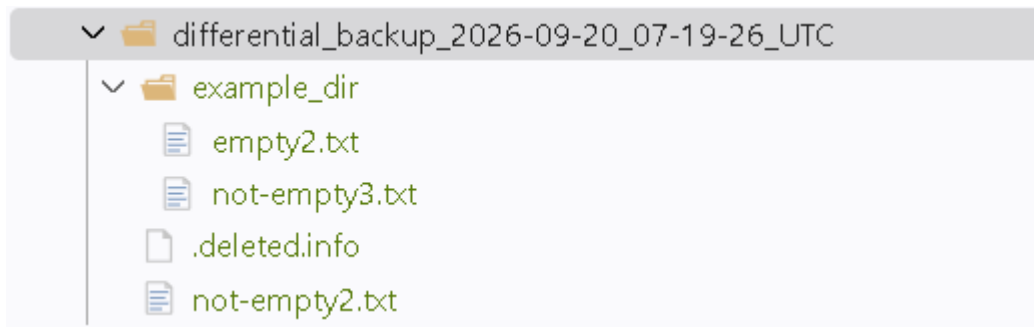
Output:

```
wilson@DESKTOP-KSAB409:/mnt/c/College_Computer_Science/Sem 7/high-performance-computing/assignment-1$ ./backup-script.sh
Removing existing BACKUP_FOLDER to avoid dup/corrupted backup
BACKUP INTERVAL: 1 minute
Running initial full backup!
Run differential backup on 2026-09-20_07-19-26_UTC
Run differential backup on 2026-09-20_07-20-26_UTC
```

full-backup untuk inisialisasi:

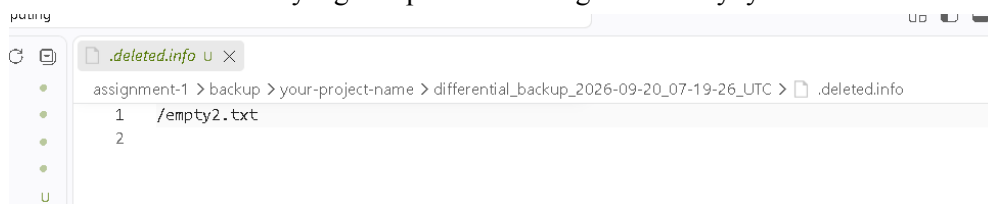
```
full-backup_2026-09-20_07-18-26_UTC
├── example_dir
│   ├── empty3.txt
│   ├── not-empty3.txt
│   ├── empty.txt
│   ├── empty2.txt
│   ├── not-empty.txt
│   └── not-empty2.txt
```

differential backup pertama:



pada bagian ini, terjadi perubahan isi file pada **not-empty2.txt** dan **non-empty3.txt** dan terjadi penambahan file baru pada **example_dir/** dan penghapusan **empty2.txt** pada **root**

.deleted.info berisi file yang dihapus sesuai dengan directorynya:



differential backup kedua:

hal yang sama dibackup karena tidak ada perubahan

